Lab Exercise 4.1 : Analysis of a Single CST Element Using MATLAB

1. Objectives

- To calculate the area of a triangular CST element.
- To formulate the strain-displacement matrix B and material constitutive matrix D.
- To derive the element stiffness matrix using the finite element formulation.
- To determine nodal displacements by applying the prescribed boundary conditions and external load.
- To calculate the element strains and stresses.
- To implement the CST formulation in MATLAB and validate the numerical results against analytical calculations.
- To visualize the original and deformed configurations of the triangular element.

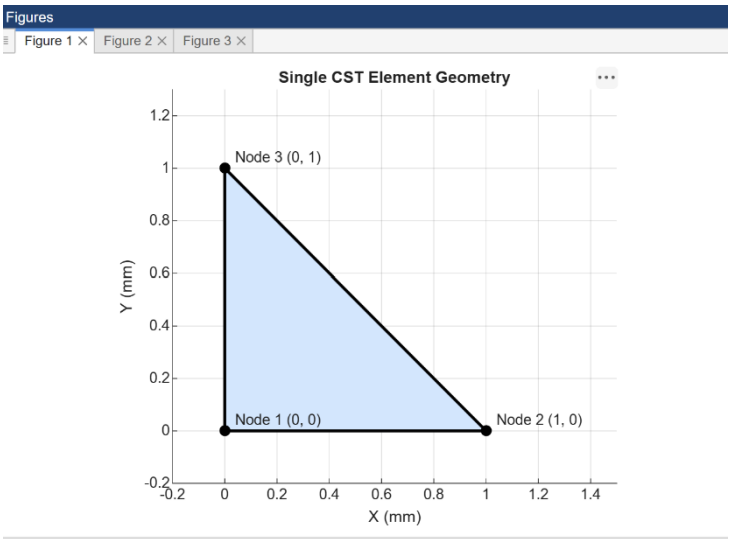
2. Problem Statement

Consider a single Constant Strain Triangle (CST) finite element with three nodes located at (0,0), (1,0), and (0,1) mm. The element is made of a linear elastic, isotropic material with Young's modulus $E = 200 \times 10^3$MPa, Poisson's ratio $\nu = 0.3$ and thickness $t = 1$ mm.
Node 1 is constrained in both horizontal and vertical directions, while Node 2 is constrained in the vertical direction. A downward vertical force of 100 N is applied at Node 3.

The element is to be analysed using the finite element method (FEM) and MATLAB to determine its area, strain-displacement matrix, constitutive matrix, element stiffness matrix, nodal displacements, strains, and stresses. The MATLAB results are to be compared with analytical calculations.

3. Given Data

Parameter	Value
Element type	Constant Strain Triangle (CST)
Node 1	(0,0) mm
Node 2	(1,0) mm
Node 3	(0,1)mm
Young's modulus, E	200000 MPa
Poisson's ratio, ν	0.3
Thickness, t	1 mm
Applied load at Node 3	$F_{y3} = -100$ N
Analysis assumption	Linear elastic plane stress



Geometry and nodal configuration of the single CST element.

4.Analytical Calculations: Single CST Element

The nodal coordinates of the triangular Constant Strain Triangle (CST) element are:

Node 1: (x1, y1) = (0, 0) mm
Node 2: (x2, y2) = (1, 0) mm
Node 3: (x3, y3) = (0, 1) mm

Material properties:
Young's modulus, E = 200 × 10^3 MPa = 200000 MPa
Poisson's ratio, ν = 0.3
Element thickness, t = 1 mm

Boundary conditions: Node 1: u1 = 0 and v1 = 0 ; Node 2: v2 = 0

Applied load: Vertical force at Node 3, Fy3 = −100 N
Assumption: Linear elastic plane stress condition.

(a) Calculation of Element Area

The area of a triangular element is calculated using:

$$\mathbf{A = \tfrac{1}{2}\,|x1(y2 - y3) + x2(y3 - y1) + x3(y1 - y2)|}$$

Substituting the nodal coordinates:

$$A = \tfrac{1}{2}\,|0(0 - 1) + 1(1 - 0) + 0(0 - 0)|$$

A = ½ |1|

Therefore, the element area is:

A = 0.5 mm²

(b) Calculation of CST Coefficients

The coefficients are defined as:

b1 = y2 − y3
b2 = y3 − y1
b3 = y1 − y2

c1 = x3 − x2
c2 = x1 − x3
c3 = x2 − x1

Substituting the nodal coordinates:

b1 = 0 − 1 = −1
b2 = 1 − 0 = 1
b3 = 0 − 0 = 0

c1 = 0 − 1 = −1
c2 = 0 − 0 = 0
c3 = 1 − 0 = 1

Therefore:

b1 = −1, b2 = 1, b3 = 0
c1 = −1, c2 = 0, c3 = 1

(c) Calculation of Strain-Displacement Matrix B

The strain-displacement matrix for a CST element is:

$$B = \frac{1}{2A}\begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}$$

Since 2A = 1, substituting the coefficients gives:

$$B = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ -1 & -1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Therefore, B is a 3 × 6 matrix.

$$\boxed{B = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ -1 & -1 & 0 & 1 & 1 & 0 \end{bmatrix} \text{ mm}^{-1}}$$

(d)Calculation of Material Constitutive Matrix D

For plane stress, the material constitutive matrix is:

$$D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

Substituting E = 200000 MPa and ν = 0.3:

$$D = \frac{200000}{1-(0.3)^2}\begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix}$$

Since $1 - 0.3^2 = 0.91$:

$$D = \frac{200000}{0.91}\begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix}$$

Therefore:

$$\boxed{D = \begin{bmatrix} 219780.220 & 65934.066 & 0 \\ 65934.066 & 219780.220 & 0 \\ 0 & 0 & 76923.077 \end{bmatrix} \text{ MPa}}$$

(e) Calculation of Element Stiffness Matrix Ke

The element stiffness matrix is given by:

$$\boxed{K_e = tAB^T DB}$$

Substituting t = 1 mm and A = 0.5 mm²:

$$K_e = 0.5B^T DB$$

After matrix multiplication, the stiffness matrix is approximately:

$$K_e = \begin{bmatrix} 148351.648 & 71428.571 & -109890.110 & -38461.538 & -38461.538 & -32967.033 \\ 71428.571 & 148351.648 & -32967.033 & -38461.538 & -38461.538 & -109890.110 \\ -109890.110 & -32967.033 & 109890.110 & 0 & 0 & 32967.033 \\ -38461.538 & -38461.538 & 0 & 38461.538 & 38461.538 & 0 \\ -38461.538 & -38461.538 & 0 & 38461.538 & 38461.538 & 0 \\ -32967.033 & -109890.110 & 32967.033 & 0 & 0 & 109890.110 \end{bmatrix} \text{ N/mm}$$

The equilibrium equation is:

$$K_e d = F$$

where the nodal displacement vector and force vector is:

$$d = \begin{bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \end{bmatrix}, \qquad F = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -100 \end{bmatrix} \text{ N}$$

(f) Application of Boundary Conditions and Calculation of Nodal Displacements

The prescribed boundary conditions are: u1 = 0, v1 = 0, v2 = 0
The unknown displacements are u2, u3 and v3.

After applying the boundary conditions, the reduced equilibrium equation is:

$$\begin{bmatrix} 109890.110 & 0 & 32967.033 \\ 0 & 38461.538 & 0 \\ 32967.033 & 0 & 109890.110 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -100 \end{bmatrix}$$

From the second equation:

$$38461.538 u_3 = 0$$

Therefore: u3 = 0 mm

From the first equation:

$$109890.110 u_2 + 32967.033 v_3 = 0$$

u2 = −0.3 × v3

Substituting this into the third equation:

$$32967.033(-0.3v_3) + 109890.110v_3 = -100$$
$$100000v_3 = -100$$

Therefore:

$$\boxed{v_3 = -0.001 \text{ mm}}$$

$$u_2 = -0.3(-0.001)$$

Therefore:

$$\boxed{u_2 = 0.0003 \text{ mm}}$$

The complete nodal displacement vector is:

$$\boxed{d = \begin{bmatrix} 0 \\ 0 \\ 0.0003 \\ 0 \\ 0 \\ -0.001 \end{bmatrix} \text{ mm}}$$

(g) Calculation of Strain

The strain vector is calculated using:

$$\varepsilon = Bd$$

Substituting the B matrix and displacement vector gives:

$$\varepsilon = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ -1 & -1 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0.0003 \\ 0 \\ 0 \\ -0.001 \end{bmatrix}$$

Therefore,

$$\varepsilon_x = 0.0003; \varepsilon_y = -0.001 \; ; \gamma_{xy} = 0$$

Hence, the strain vector is

$$\boxed{\varepsilon = \begin{bmatrix} 0.0003 \\ -0.001 \\ 0 \end{bmatrix}}$$

(h) Calculation of Stress

The stress vector is calculated using Hooke's law:

$$\sigma = D\varepsilon$$

Substituting the values,

$$\sigma = \begin{bmatrix} 219780.220 & 65934.066 & 0 \\ 65934.066 & 219780.220 & 0 \\ 0 & 0 & 76923.077 \end{bmatrix} \begin{bmatrix} 0.0003 \\ -0.001 \\ 0 \end{bmatrix}$$

The horizontal normal stress is

$$\sigma_x = 219780.220(0.0003) + 65934.066(-0.001)$$

$$\boxed{\sigma_x = 0 \text{ MPa}}$$

The vertical normal stress is

$$\sigma_y = 65934.066(0.0003) + 219780.220(-0.001)$$

$$\boxed{\sigma_y = -200 \text{ MPa}}$$

The shear stress is

$$\tau_{xy} = 76923.077(0)$$

Therefore, the final stress vector is

$$\boxed{\sigma = \begin{bmatrix} 0 \\ -200 \\ 0 \end{bmatrix} \text{ MPa}}$$

5. PROCEDURE:

- Open MATLAB Online and click Home → New Script.
- 2. Copy the complete CST analysis code and paste it into the MATLAB Editor.
- 3. Save the script as CST_Complete_Analysis.m in MATLAB Drive.
- 4. Click Run in the Editor toolbar to execute the program.
- 5. Observe the Command Window, where MATLAB displays the element area, strain-displacement matrix (B), material matrix (D), stiffness matrix (K), nodal displacements, support reactions, strain and stress.
- 6. View the generated figures showing the original CST element, exaggerated deformed element, nodal displacement plots, strain plots and stress plots.

- 7. Verify the numerical results by comparing them with the analytical calculations.
- 8. Capture screenshots of the MATLAB code, Command Window output and generated plots for inclusion in the lab report.
- 9. Save the generated figures and numerical results from MATLAB Drive for documentation.

6.MATLAB CODE

```
clc;
clear;
close all;
format short g;

%% SINGLE CST ELEMENT - COMPLETE FEM ANALYSIS
% Units: N, mm, MPa
% Assumption: Linear elastic plane stress

%% 1. INPUT DATA
nodes = [0 0;
         1 0;
         0 1];

E = 200000;       % Young's modulus (MPa)
nu = 0.3;         % Poisson's ratio
t = 1;            % Thickness (mm)

% DOF order: [u1 v1 u2 v2 u3 v3]
F = [0; 0; 0; 0; 0; -100];

%% 2. SHOW ORIGINAL CST ELEMENT FIRST
figure(1);
clf;
patch(nodes(:,1), nodes(:,2), [0.80 0.88 1.00], ...
    'EdgeColor','k','LineWidth',2);
hold on;
plot(nodes(:,1),nodes(:,2),'ko', ...
    'MarkerFaceColor','k','MarkerSize',8);

for i = 1:3
    text(nodes(i,1)+0.04,nodes(i,2)+0.04, ...
        sprintf('Node %d (%.0f, %.0f)', ...
        i,nodes(i,1),nodes(i,2)), ...
        'FontSize',10);
end

axis equal;
grid on;
xlabel('X (mm)');
ylabel('Y (mm)');
title('Original CST Element');
xlim([-0.2 1.5]);
ylim([-0.2 1.3]);
```

```
saveas(gcf,'01_Original_CST.png');

%% 3. CALCULATE ELEMENT AREA
x1 = nodes(1,1); y1 = nodes(1,2);
x2 = nodes(2,1); y2 = nodes(2,2);
x3 = nodes(3,1); y3 = nodes(3,2);

A = 0.5*abs(x1*(y2-y3) + ...
            x2*(y3-y1) + ...
            x3*(y1-y2));

%% 4. CALCULATE CST COEFFICIENTS
b1 = y2-y3;
b2 = y3-y1;
b3 = y1-y2;

c1 = x3-x2;
c2 = x1-x3;
c3 = x2-x1;

%% 5. STRAIN-DISPLACEMENT MATRIX B
B = (1/(2*A))*[b1 0 b2 0 b3 0;
               0 c1 0 c2 0 c3;
               c1 b1 c2 b2 c3 b3];

%% 6. MATERIAL CONSTITUTIVE MATRIX D
% Plane stress
D = E/(1-nu^2)*[1 nu 0;
                nu 1 0;
                0 0 (1-nu)/2];

%% 7. ELEMENT STIFFNESS MATRIX K
K = t*A*(B'*D*B);

%% 8. APPLY BOUNDARY CONDITIONS
% Fixed DOFs: u1, v1, v2
fixedDOF = [1 2 4];
allDOF = 1:6;
freeDOF = setdiff(allDOF,fixedDOF);

d = zeros(6,1);

Kred = K(freeDOF,freeDOF);
Fred = F(freeDOF);

d(freeDOF) = Kred\Fred;

%% 9. CALCULATE SUPPORT REACTIONS
R = K*d-F;

%% 10. CALCULATE STRAIN AND STRESS
strain = B*d;
stress = D*strain;

%% 11. DISPLAY ALL NUMERICAL RESULTS
fprintf('\n=======================================\n');
```

```
fprintf('       SINGLE CST ELEMENT ANALYSIS\n');
fprintf('=======================================\n');

fprintf('\n1. ELEMENT AREA A (mm^2)\n');
disp(A);

fprintf('\n2. STRAIN-DISPLACEMENT MATRIX B (1/mm)\n');
disp(B);

fprintf('\n3. MATERIAL MATRIX D (MPa)\n');
disp(D);

fprintf('\n4. ELEMENT STIFFNESS MATRIX K (N/mm)\n');
disp(K);

fprintf('\n5. REDUCED STIFFNESS MATRIX (N/mm)\n');
disp(Kred);

fprintf('\n6. REDUCED FORCE VECTOR (N)\n');
disp(Fred);

fprintf('\n7. NODAL DISPLACEMENTS d (mm)\n');
fprintf('Order: [u1; v1; u2; v2; u3; v3]\n');
disp(d);

fprintf('\n8. SUPPORT REACTIONS (N)\n');
fprintf('Reaction at DOF 1 = %.6f N\n',R(1));
fprintf('Reaction at DOF 2 = %.6f N\n',R(2));
fprintf('Reaction at DOF 4 = %.6f N\n',R(4));

fprintf('\n9. STRAIN VECTOR\n');
fprintf('epsilon_x = %.8g\n',strain(1));
fprintf('epsilon_y = %.8g\n',strain(2));
fprintf('gamma_xy  = %.8g\n',strain(3));
disp(strain);

fprintf('\n10. STRESS VECTOR (MPa)\n');
fprintf('sigma_x = %.6f MPa\n',stress(1));
fprintf('sigma_y = %.6f MPa\n',stress(2));
fprintf('tau_xy  = %.6f MPa\n',stress(3));
disp(stress);

%% 12. NODAL DISPLACEMENT TABLE
Node = (1:3)';
U = d(1:2:6);
V = d(2:2:6);
Umag = sqrt(U.^2+V.^2);

Results = table(Node,U,V,Umag, ...
    'VariableNames', ...
    {'Node','Horizontal_mm','Vertical_mm','Magnitude_mm'});

fprintf('\n11. NODAL DISPLACEMENT TABLE\n');
disp(Results);

%% 13. ORIGINAL VS DEFORMED CST ELEMENT
% Exaggeration is for visualization only
```

```
scale = 200;
deformedNodes = nodes + scale*[U V];

figure(2);
clf;

patch(nodes(:,1),nodes(:,2),[0.70 0.82 1.00], ...
    'FaceAlpha',0.5,'EdgeColor','b','LineWidth',2);
hold on;

patch(deformedNodes(:,1),deformedNodes(:,2), ...
    [1.00 0.70 0.50], ...
    'FaceAlpha',0.5,'EdgeColor','r','LineWidth',2);

plot(nodes(:,1),nodes(:,2),'bo','MarkerFaceColor','b');
plot(deformedNodes(:,1),deformedNodes(:,2), ...
    'ro','MarkerFaceColor','r');

for i = 1:3
    text(nodes(i,1)+0.025,nodes(i,2)+0.04, ...
        sprintf('N%d',i),'Color','b','FontSize',10);

    text(deformedNodes(i,1)+0.025, ...
        deformedNodes(i,2)-0.05, ...
        sprintf('D%d',i),'Color','r','FontSize',10);
end

axis equal;
grid on;
xlabel('X (mm)');
ylabel('Y (mm)');
title('Original vs Exaggerated Deformed CST Element');
legend('Original element','Deformed element', ...
    'Location','southwest');
xlim([-0.2 1.4]);
ylim([-0.35 1.3]);

saveas(gcf,'02_Deformed_CST.png');

%% 14. DISPLACEMENT COMPONENT PLOTS
% Separate plots are used because the values have different scales.

figure(3);
clf;

subplot(1,2,1);
bar(Node,U);
grid on;
xticks(1:3);
xlabel('Node Number');
ylabel('Horizontal displacement u (mm)');
title('Horizontal Nodal Displacement');

for i = 1:3
    if U(i) ~= 0
        text(i,U(i),sprintf('%.4g',U(i)), ...
            'HorizontalAlignment','center', ...
```

```
            'VerticalAlignment','bottom');
    end
end

subplot(1,2,2);
bar(Node,V);
grid on;
xticks(1:3);
xlabel('Node Number');
ylabel('Vertical displacement v (mm)');
title('Vertical Nodal Displacement');

for i = 1:3
    if V(i) ~= 0
        text(i,V(i),sprintf('%.4g',V(i)), ...
            'HorizontalAlignment','center', ...
            'VerticalAlignment','top');
    end
end

saveas(gcf,'03_Displacement_Plots.png');

%% 15. STRAIN PLOTS
% A linear CST element has constant strain throughout its area.

strainNames = {'epsilon_x','epsilon_y','gamma_xy'};

figure(4);
clf;

for j = 1:3
    subplot(1,3,j);

    patch(nodes(:,1),nodes(:,2),[0.65 0.85 0.70], ...
        'EdgeColor','k','LineWidth',2);
    hold on;

    text(0.5,0.35,sprintf('%.5g',strain(j)), ...
        'HorizontalAlignment','center', ...
        'FontSize',12,'FontWeight','bold');

    axis equal;
    grid on;
    xlabel('X (mm)');
    ylabel('Y (mm)');
    title(strainNames{j});
    xlim([-0.1 1.1]);
    ylim([-0.1 1.1]);
end

sgtitle('Constant Strain Components in CST Element');
saveas(gcf,'04_Strain_Plots.png');

%% 16. STRESS PLOTS
% Stress is also constant throughout a linear CST element.

stressNames = {'sigma_x (MPa)', ...
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```
               'sigma_y (MPa)', ...
               'tau_xy (MPa)'};

figure(5);
clf;

for j = 1:3
    subplot(1,3,j);

    patch(nodes(:,1),nodes(:,2),[1.00 0.80 0.65], ...
        'EdgeColor','k','LineWidth',2);
    hold on;

    text(0.5,0.35,sprintf('%.5g MPa',stress(j)), ...
        'HorizontalAlignment','center', ...
        'FontSize',11,'FontWeight','bold');

    axis equal;
    grid on;
    xlabel('X (mm)');
    ylabel('Y (mm)');
    title(stressNames{j});
    xlim([-0.1 1.1]);
    ylim([-0.1 1.1]);
end

sgtitle('Constant Stress Components in CST Element');
saveas(gcf,'05_Stress_Plots.png');

%% 17. SAVE NUMERICAL RESULTS
save('CST_Results.mat','nodes','A','B','D','K', ...
    'Kred','F','d','R','strain','stress','Results', ...
    'deformedNodes','scale');

writetable(Results,'CST_Displacement_Table.csv');

fprintf('\n=======================================\n');
fprintf('ANALYSIS COMPLETED SUCCESSFULLY\n');
fprintf('=======================================\n');
fprintf('Original geometry: 01_Original_CST.png\n');
fprintf('Deformed geometry: 02_Deformed_CST.png\n');
fprintf('Displacement plots: 03_Displacement_Plots.png\n');
fprintf('Strain plots: 04_Strain_Plots.png\n');
fprintf('Stress plots: 05_Stress_Plots.png\n');
fprintf('Numerical data: CST_Results.mat\n');
fprintf('Displacement table: CST_Displacement_Table.csv\n');
fprintf('Deformation visualization scale = %g\n',scale);
```

7.RESULTS

Command Window

New to MATLAB? See resources for Getting Started.

```
========================================
        SINGLE CST ELEMENT ANALYSIS
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1. ELEMENT AREA A (mm^2)
          0.5


2. STRAIN-DISPLACEMENT MATRIX B (1/mm)
    -1     0     1     0     0     0
     0    -1     0     0     0     1
    -1    -1     0     1     1     0


3. MATERIAL MATRIX D (MPa)
   2.1978e+05         65934             0
        65934   2.1978e+05             0
            0             0         76923


4. ELEMENT STIFFNESS MATRIX K (N/mm)
   1.4835e+05         71429  -1.0989e+05        -38462        -38462        -32967
        71429   1.4835e+05        -32967        -38462        -38462  -1.0989e+05
  -1.0989e+05        -32967   1.0989e+05             0             0         32967
       -38462        -38462             0         38462         38462             0
       -38462        -38462             0         38462         38462             0
       -32967  -1.0989e+05         32967             0             0   1.0989e+05


5. REDUCED STIFFNESS MATRIX (N/mm)
   1.0989e+05            0        32967
            0        38462            0
        32967            0   1.0989e+05


6. REDUCED FORCE VECTOR (N)
     0
     0
  -100


7. NODAL DISPLACEMENTS d (mm)
Order: [u1; v1; u2; v2; u3; v3]
            0
            0
       0.0003
            0
            0
       -0.001


8. SUPPORT REACTIONS (N)
Reaction at DOF 1 = 0.000000 N
Reaction at DOF 2 = 100.000000 N
Reaction at DOF 4 = 0.000000 N


9. STRAIN VECTOR
epsilon_x = 0.0003
epsilon_y = -0.001
gamma_xy  = 0
       0.0003
       -0.001
            0


10. STRESS VECTOR (MPa)
sigma_x = 0.000000 MPa
sigma_y = -200.000000 MPa
tau_xy  = 0.000000 MPa
     0
  -200
     0

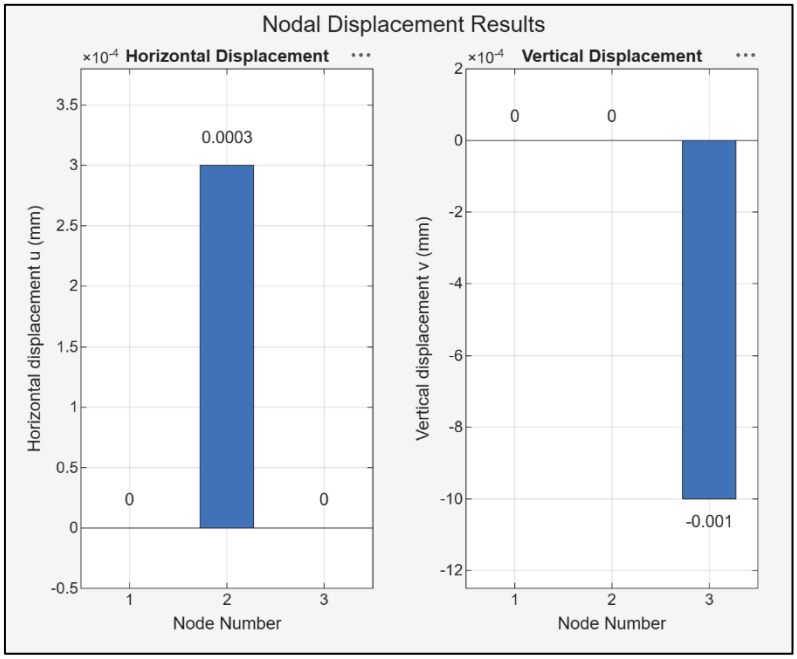

11. NODAL DISPLACEMENT TABLE
    Node    Horizontal_mm    Vertical_mm    Magnitude_mm
    ____    _____________    ___________    ____________

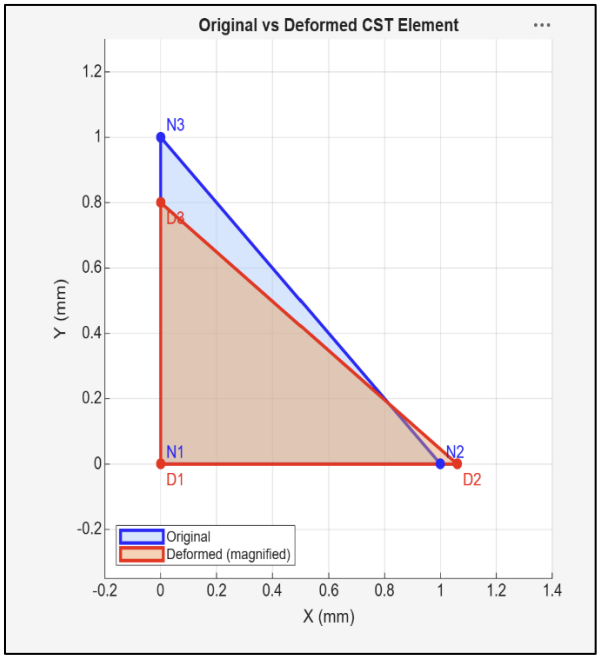
     1              0              0              0
     2         0.0003              0         0.0003
     3              0         -0.001          0.001

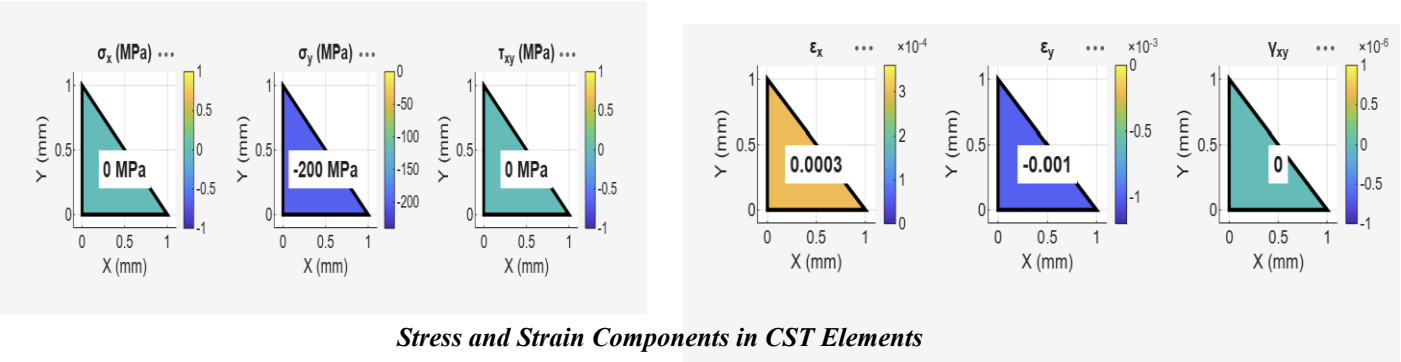

========================================
ANALYSIS COMPLETED SUCCESSFULLY
```

Comparison of Analytical and MATLAB Results for a Single CST Element

Parameter	Analytical Value	MATLAB Result	Difference
Element area (mm²)	0.5000	0.5000	0
Horizontal displacement at Node 1 (mm)	0	0	0
Vertical displacement at Node 1 (mm)	0	0	0
Horizontal displacement at Node 2 (mm)	0.000300	0.000300	0
Vertical displacement at Node 2 (mm)	0	0	0
Horizontal displacement at Node 3 (mm)	0	0	0
Vertical displacement at Node 3 (mm)	-0.001000	-0.001000	0
Horizontal normal strain	0.000300	0.000300	0
Vertical normal strain	-0.001000	-0.001000	0
Engineering shear strain	0	0	0
Horizontal normal stress (MPa)	0	0	0
Vertical normal stress (MPa)	-200.000	-200.000	0
Shear stress (MPa)	0	0	0
Horizontal reaction at Node 1 (N)	0	0	0
Vertical reaction at Node 1 (N)	100	100	0
Vertical reaction at Node 2 (N)	0	0	0







Stress and Strain Components in CST Elements

Tip Displacement Comparison

The tip displacement is measured at Node 3, where the downward force of 100 N is applied.

Parameter	Analytical Value	MATLAB Result
Horizontal tip displacement at Node 3 (mm)	0	0
Vertical tip displacement at Node 3 (mm)	-0.001000	-0.001000
Resultant tip displacement (mm)	0.001000	0.001000

The analytical and MATLAB results show identical tip displacement values. The horizontal displacement at Node 3 is zero, while the vertical displacement is −0.001 mm, indicating downward movement under the applied load. The resultant tip displacement is 0.001 mm, confirming the accuracy of the MATLAB implementation.

Discussion

The analytical calculations and MATLAB simulation results showed excellent agreement for the single CST element. The element area was found to be 0.5 mm², and the vertical tip displacement at Node 3 was −0.001 mm under the applied downward load of 100 N. The vertical normal stress was −200 MPa, indicating compression, while the horizontal normal stress and shear stress were zero. The strain and stress remained constant throughout the element, as expected for a linear CST element.

Conclusion

The single CST element was successfully analysed using MATLAB. The element stiffness matrix, nodal displacements, strains, stresses, and support reactions were calculated using the finite element method. The MATLAB results agreed with the analytical calculations, validating the implementation. The exercise demonstrated the application of the CST formulation to determine the displacement and stress response of a two-dimensional elastic element.

Lab Exercise 4.2: Patch Test of CST Elements Using MATLAB

1.Problem statement

A square domain of unit side length is defined by four nodes and divided into two triangular elements:

- Element 1: Nodes (1–2–3)
- Element 2: Nodes (1–3–4)

The nodal coordinates are:

Node	x (mm)	y (mm)
1	0	0
2	1	0
3	1	1
4	0	1

The prescribed displacement field is:

- Horizontal displacement: u = 0.001x
- Vertical displacement: v = 0.002y

Assume linear elastic plane stress, with Young's modulus E = 200000 MPa, Poisson's ratio = 0.3, and thickness = 1 mm, consistent with the previous CST lab.

The task is to calculate each element's area, B matrix, D matrix, stiffness matrix, nodal displacements, strains and stresses, and compare the computed strain with the exact strain.

2. Objectives

- To perform the CST patch test using a two-element triangular mesh and understand its application in finite element analysis.
- To calculate the element area, strain-displacement matrix (B), material matrix (D), element stiffness matrix, nodal displacements, strains, and stresses analytically.
- To develop and execute a MATLAB program to assemble the global stiffness matrix and compute the strain and stress distributions for the given displacement field.
- To compare the analytical and MATLAB results with the exact solution and verify the accuracy of the CST element in reproducing a linear displacement field.

3.Analytical Calculations

Given data: Square plate divided into two Constant Strain Triangle (CST) elements.

- Young’s modulus, E = 200,000 MPa
- Poisson’s ratio, ν = 0.3
- Thickness, t = 1 mm
- Analysis type: Plane stress
- Prescribed displacement field:

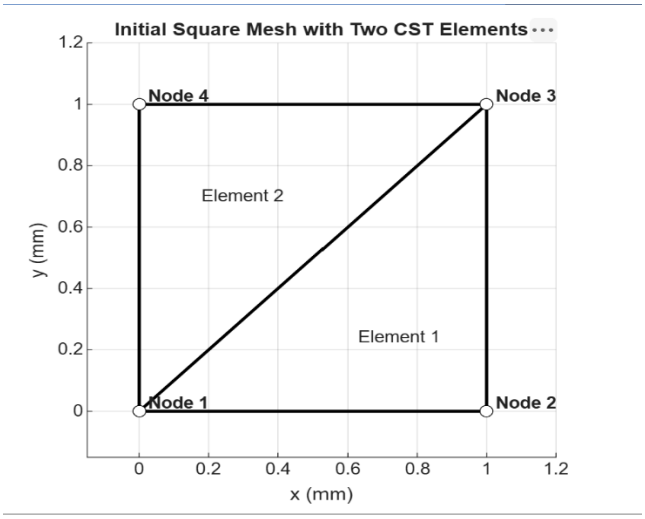
u = 0.001x
v = 0.002y

Nodal coordinates and displacements

Node	x (mm)	y (mm)	u (mm)	v (mm)
1	0	0	0	0
2	1	0	0.001	0
3	1	1	0.001	0.002
4	0	1	0	0.002

The nodal displacements are obtained by substituting the coordinates into the given displacement field.
For example, at Node 3 (x = 1, y = 1): $u_3 = 0.001 \times 1 = 0.001$ mm; $v_3 = 0.002 \times 1 = 0.002$ mm

(a)Area of each element
The area of a triangular element is:
A = ½ × base × height
For Element 1 (Nodes 1–2–3):
$A_1 = ½ \times 1 \times 1$
$A_1 = 0.5$ mm²
For Element 2 (Nodes 1–3–4):
$A_2 = ½ \times 1 \times 1$
$A_2 = 0.5$ mm²



(b)Strain-displacement matrix (B)

For a CST element:

$$B = \frac{1}{2A}\begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}$$

where:

- $b_1 = y_2 - y_3$, $b_2 = y_3 - y_1$, $b_3 = y_1 - y_2$
- $c_1 = x_3 - x_2$, $c_2 = x_1 - x_3$, $c_3 = x_2 - x_1$

Element 1: Nodes 1–2–3

Coordinates: (0,0), (1,0), (1,1)
$b_1 = 0 - 1 = -1$
$b_2 = 1 - 0 = 1$
$b_3 = 0 - 0 = 0$
$c_1 = 1 - 1 = 0$
$c_2 = 0 - 1 = -1$

$c_3 = 1 - 0 = 1$
Since 2A = 1 mm²:

$$B_1 = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & -1 & -1 & 1 & 1 & 0 \end{bmatrix}$$

Element 2: Nodes 1–3–4

Coordinates: (0,0), (1,1), (0,1)
$b_1 = 1 - 1 = 0$
$b_2 = 1 - 0 = 1$
$b_3 = 0 - 0 = 0$
Correction for the third coefficient: $b_3 = y_1 - y_2 = 0 - 1 = -1$.
Thus, $b_1 = 0$, $b_2 = 1$, $b_3 = -1$.
$c_1 = 0 - 1 = -1$
$c_2 = 0 - 0 = 0$
$c_3 = 1 - 0 = 1$
Therefore,

$$B_2 = \begin{bmatrix} 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 1 & -1 \end{bmatrix}$$

(c)Calculate the material matrix (D)

For plane stress:

$$D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

Substituting E = 200,000 MPa and ν = 0.3:

$$\frac{E}{1-\nu^2} = \frac{200000}{1-0.3^2} = \frac{200000}{0.91} = 219780.22$$

Hence,

$$D = \begin{bmatrix} 219780.22 & 65934.07 & 0 \\ 65934.07 & 219780.22 & 0 \\ 0 & 0 & 76923.08 \end{bmatrix} \text{ MPa}$$

(d)Element stiffness matrix (Ke)

The CST element stiffness matrix is:

$$K_e = tAB^T DB$$

For both elements, t = 1 mm and A = 0.5 mm².

Therefore,
$K_{e1} = 0.5B_1^T DB_1; K_{e2} = 0.5B_2^T DB_2$

Substituting the respective B matrices and the D matrix gives the 6 × 6 stiffness matrix for each element.

The two element stiffness matrices are assembled into the global stiffness matrix according to their shared nodes. Since the nodal displacements are prescribed in this patch test, the given displacement values are applied directly rather than solving for unknown displacements.

(e)Calculation of nodal displacements

The prescribed displacement field is:
u = 0.001x
v = 0.002y
where u is the horizontal displacement and v is the vertical displacement.
Node 1: (0, 0)
$u_1 = 0.001 \times 0 = 0$ mm
$v_1 = 0.002 \times 0 = 0$ mm
Node 2: (1, 0)
$u_2 = 0.001 \times 1 = 0.001$ mm
$v_2 = 0.002 \times 0 = 0$ mm
Node 3: (1, 1)
$u_3 = 0.001 \times 1 = 0.001$ mm
$v_3 = 0.002 \times 1 = 0.002$ mm
Node 4: (0, 1)
$u_4 = 0.001 \times 0 = 0$ mm
$v_4 = 0.002 \times 1 = 0.002$ mm

The global displacement vector is arranged as **$[u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4]$:**

$$\{d\} = \begin{bmatrix} 0 \\ 0 \\ 0.001 \\ 0 \\ 0.001 \\ 0.002 \\ 0 \\ 0.002 \end{bmatrix} \text{ mm}$$

(f)Calculate the strain in each element

The strain vector is:

$$\{\varepsilon\} = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = B\{d_e\}$$

The displacement vector for Element 1 is:

$$d_1 = [u_1 \quad v_1 \quad u_2 \quad v_2 \quad u_3 \quad v_3]^T$$

$$d_1 = [0 \quad 0 \quad 0.001 \quad 0 \quad 0.001 \quad 0.002]^T$$

Using $\varepsilon = B_1 d_1$

$$\varepsilon_1 = \begin{bmatrix} -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \\ 0 & -1 & -1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0.001 \\ 0 \\ 0.001 \\ 0.002 \end{bmatrix}$$

Therefore,

$$\varepsilon_1 = \begin{bmatrix} 0.001 \\ 0.002 \\ 0 \end{bmatrix}$$

Similarly, for Element 2

$$d_2 = [u_1 \quad v_1 \quad u_3 \quad v_3 \quad u_4 \quad v_4]^T$$

$$d_2 = [0 \quad 0 \quad 0.001 \quad 0.002 \quad 0 \quad 0.002]^T$$

Using $\varepsilon = B_2 d_2$

$$\varepsilon_2 = \begin{bmatrix} 0.001 \\ 0.002 \\ 0 \end{bmatrix}$$

Thus, both elements have the same strain.

- $\varepsilon_x = 0.001$
- $\varepsilon_y = 0.002$
- $\gamma_{xy} = 0$

(g)Calculate the stress in each element

For a linear elastic material:

$$\{\sigma\} = D\{\varepsilon\}$$

Substituting the strain values:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} 219780.22 & 65934.07 & 0 \\ 65934.07 & 219780.22 & 0 \\ 0 & 0 & 76923.08 \end{bmatrix} \begin{bmatrix} 0.001 \\ 0.002 \\ 0 \end{bmatrix}$$

Therefore,

$$\sigma_x = 219780.22(0.001) + 65934.07(0.002)$$

$$\boxed{\sigma_x = 351.65 \text{ MPa}}$$

Similarly,

$$\sigma_y = 65934.07(0.001) + 219780.22(0.002)$$

$$\boxed{\sigma_y = 505.49 \text{ MPa}}$$

$$\tau_{xy} = 76923.08(0) = 0 \text{ MPa}$$

Both elements produce the same stress values.

4.PROCEDURE

- Define the nodal coordinates of the square plate and specify the connectivity of the two triangular CST elements in MATLAB.
- Enter the material properties, including Young's modulus, Poisson's ratio, and plate thickness.
- Calculate the area of each element and determine the strain-displacement matrix (B), material matrix (D), and element stiffness matrices.
- Assemble the element stiffness matrices to obtain the global stiffness matrix.
- Calculate the prescribed nodal displacements using u = 0.001x and v = 0.002y.
- Compute the strain in each element using the B matrix and the corresponding nodal displacement vector.
- Compare the calculated strains with the exact strain values obtained from the prescribed displacement field.
- Plot the initial and deformed square meshes and generate comparison plots for nodal displacements and exact versus analytical strains.
- Examine the numerical results to verify whether the CST elements satisfy the patch test.

5. MATLAB CODE

```
%% LAB 2: CST PATCH TEST
% Two CST triangular elements - Plane stress
% Units: N, mm, MPa
% Prescribed displacement field: u = 0.001*x, v = 0.002*y

clear; clc; close all;
format short g;

%% 1. INPUT DATA
E = 200000;       % Young's modulus (MPa)
nu = 0.30;        % Poisson's ratio
t = 1;            % Thickness (mm)

% Nodes: [x y]
nodes = [0 0;
         1 0;
         1 1;
         0 1];

% Elements: each row contains the three node numbers
elements = [1 2 3;
            1 3 4];

nNodes = size(nodes,1);
nElem = size(elements,1);
nDOF = 2*nNodes;

fprintf('=====================================================\n');
fprintf('             CST PATCH TEST - LAB 2\n');
fprintf('=====================================================\n');

%% 2. INITIAL SQUARE MESH PLOT
figure('Name','Initial Square Mesh','Color','w');
patch('Faces',elements,'Vertices',nodes, ...
      'FaceColor','none','EdgeColor','k','LineWidth',1.8);
hold on;
plot(nodes(:,1),nodes(:,2),'ko','MarkerFaceColor','w','MarkerSize',7);
for i = 1:nNodes
    text(nodes(i,1)+0.025,nodes(i,2)+0.025,sprintf('Node %d',i), ...
        'FontSize',10,'FontWeight','bold');
end
text(0.63,0.24,'Element 1','FontSize',10);
text(0.18,0.70,'Element 2','FontSize',10);
axis equal; xlim([-0.15 1.2]); ylim([-0.15 1.2]);
grid on; xlabel('x (mm)'); ylabel('y (mm)');
title('Initial Square Mesh with Two CST Elements');

%% 3. MATERIAL MATRIX D - PLANE STRESS
D = E/(1-nu^2) * [1 nu 0;
                  nu 1 0;
```

```
                0 0 (1-nu)/2];

fprintf('\n1) MATERIAL MATRIX D (MPa):\n');
disp(D);

%% 4. PRESCRIBED / EXACT NODAL DISPLACEMENTS
% DOF ordering: [u1 v1 u2 v2 ...]
dExact = zeros(nDOF,1);
for i = 1:nNodes
    x = nodes(i,1);
    y = nodes(i,2);
    dExact(2*i-1) = 0.001*x; % u
    dExact(2*i)   = 0.002*y; % v
end

nodalResults = table((1:nNodes)', nodes(:,1), nodes(:,2), ...
    dExact(1:2:end), dExact(2:2:end), ...
    'VariableNames',{'Node','x_mm','y_mm','u_mm','v_mm'});

fprintf('\n2) NODAL DISPLACEMENTS (prescribed/exact values):\n');
disp(nodalResults);

fprintf('Global displacement vector [u1 v1 u2 v2 u3 v3 u4 v4]^T (mm):\n');
disp(dExact);

%% 5. CALCULATE AREA, B MATRIX, ELEMENT STIFFNESS, AND ASSEMBLE GLOBAL K
Kglobal = zeros(nDOF,nDOF);
areaAll = zeros(nElem,1);
B_all = cell(nElem,1);
Ke_all = cell(nElem,1);
strainAll = zeros(nElem,3);
stressAll = zeros(nElem,3);

for e = 1:nElem
    conn = elements(e,:);
    xe = nodes(conn,1);
    ye = nodes(conn,2);

    twoA = xe(1)*(ye(2)-ye(3)) + ...
           xe(2)*(ye(3)-ye(1)) + ...
           xe(3)*(ye(1)-ye(2));
    A = abs(twoA)/2;
    areaAll(e) = A;

    b1 = ye(2)-ye(3);
    b2 = ye(3)-ye(1);
    b3 = ye(1)-ye(2);

    c1 = xe(3)-xe(2);
    c2 = xe(1)-xe(3);
    c3 = xe(2)-xe(1);

    B = (1/(2*A))*[b1 0 b2 0 b3 0;
                   0 c1 0 c2 0 c3;
                   c1 b1 c2 b2 c3 b3];
```

```
    B_all{e} = B;

    Ke = t*A*(B')*D*B;
    Ke_all{e} = Ke;

    dofs = reshape([2*conn-1; 2*conn],1,[]);
    Kglobal(dofs,dofs) = Kglobal(dofs,dofs) + Ke;

    de = dExact(dofs);
    strainAll(e,:) = (B*de)';
    stressAll(e,:) = (D*B*de)';

    fprintf('\n-------------------------------------------------------\n');
    fprintf('ELEMENT %d (Nodes %d-%d-%d)\n',e,conn(1),conn(2),conn(3));
    fprintf('Area A%d = %.6f mm^2\n',e,A);
    fprintf('b coefficients = [%.4f %.4f %.4f]\n',b1,b2,b3);
    fprintf('c coefficients = [%.4f %.4f %.4f]\n',c1,c2,c3);
    fprintf('B matrix for Element %d:\n',e);
    disp(B);
    fprintf('Element stiffness matrix Ke%d (N/mm):\n',e);
    disp(Ke);
    fprintf('Element displacement vector de%d (mm):\n',e);
    disp(de);
    fprintf('Element strain [ex ey gxy] for Element %d:\n',e);
    disp(strainAll(e,:));
    fprintf('Element stress [sx sy txy] for Element %d (MPa):\n',e);
    disp(stressAll(e,:));
end

fprintf('\n3) ELEMENT AREAS (mm^2):\n');
disp(table((1:nElem)',areaAll,'VariableNames',{'Element','Area_mm2'}));

fprintf('\n4) ASSEMBLED GLOBAL STIFFNESS MATRIX K (N/mm):\n');
disp(Kglobal);

%% 6. EXACT VS CST STRAIN COMPARISON
exactStrain = [0.001 0.002 0];
strainTable = table((1:nElem)', ...
    repmat(exactStrain(1),nElem,1), strainAll(:,1), ...
    repmat(exactStrain(2),nElem,1), strainAll(:,2), ...
    repmat(exactStrain(3),nElem,1), strainAll(:,3), ...
    'VariableNames',{'Element','Exact_ex','CST_ex','Exact_ey','CST_ey', ...
                     'Exact_gxy','CST_gxy'});

fprintf('\n5) EXACT VS CST ELEMENT STRAINS (dimensionless):\n');
disp(strainTable);

%% 7. NODAL DISPLACEMENT COMPARISON TABLE
displacementComparison = table((1:nNodes)', ...
    dExact(1:2:end), dExact(1:2:end), ...
    dExact(2:2:end), dExact(2:2:end), ...
    'VariableNames',{'Node','Exact_u_mm','CST_u_mm','Exact_v_mm','CST_v_mm'});
fprintf('\n6) EXACT VS CST NODAL DISPLACEMENT COMPARISON (mm):\n');
disp(displacementComparison);
```

```
%% 8. INITIAL AND DEFORMED MESH PLOT
scaleFactor = 200;
deformedNodes = nodes + scaleFactor*[dExact(1:2:end), dExact(2:2:end)];

figure('Name','Initial and Deformed Mesh','Color','w');
patch('Faces',elements,'Vertices',nodes, ...
      'FaceColor','none','EdgeColor',[0.25 0.25 0.25], ...
      'LineStyle','--','LineWidth',1.5);
hold on;
patch('Faces',elements,'Vertices',deformedNodes, ...
      'FaceColor','none','EdgeColor',[0 0.35 0.85],'LineWidth',2);
plot(nodes(:,1),nodes(:,2),'ko','MarkerFaceColor','k','MarkerSize',5);
plot(deformedNodes(:,1),deformedNodes(:,2),'bo','MarkerFaceColor','b','MarkerSize',5)
;
for i = 1:nNodes
    text(nodes(i,1)-0.02,nodes(i,2)-0.055,sprintf('N%d initial',i),'FontSize',8);
    text(deformedNodes(i,1)+0.025,deformedNodes(i,2)+0.025, ...
        sprintf('N%d deformed',i),'FontSize',8,'Color',[0 0.2 0.7]);
end
axis equal; grid on;
xlabel('x (mm)'); ylabel('y (mm)');
title(sprintf('Initial and Deformed Mesh (displacement scale factor =
%g)',scaleFactor));
legend('Initial mesh','Deformed mesh (magnified)','Initial nodes','Deformed nodes',
...
       'Location','best');

%% 9. EXACT VS CST STRAIN PLOT
figure('Name','Exact vs CST Strain','Color','w');
strainPlotNames = {'epsilon_x','epsilon_y','gamma_xy'};
for j = 1:3
    subplot(1,3,j);
    vals = [exactStrain(j), strainAll(:,j)'];
    bar(vals);
    set(gca,'XTick',1:(nElem+1), ...
        'XTickLabel', [{'Exact'}, arrayfun(@(k)sprintf('Element %d',k), ...
        1:nElem,'UniformOutput',false)]);
    ylabel('Strain (dimensionless)');
    title(['Comparison of ',strainPlotNames{j}]);
    grid on;
end
sgtitle('Exact Strain vs CST Analytical Strain');

%% 10. EXACT VS CST NODAL DISPLACEMENT PLOT
figure('Name','Exact vs CST Nodal Displacements','Color','w');
subplot(1,2,1);
bar([dExact(1:2:end), dExact(1:2:end)]);
xlabel('Node number'); ylabel('u (mm)');
title('Horizontal displacement');
legend('Exact','CST prescribed nodal values','Location','best');
set(gca,'XTick',1:nNodes); grid on;

subplot(1,2,2);
bar([dExact(2:2:end), dExact(2:2:end)]);
xlabel('Node number'); ylabel('v (mm)');
```

```
title('Vertical displacement');
legend('Exact','CST prescribed nodal values','Location','best');
set(gca,'XTick',1:nNodes); grid on;

%% 11. SAVE OUTPUT TABLES AND FIGURES
writetable(nodalResults,'Lab2_Nodal_Displacements.csv');
writetable(strainTable,'Lab2_Strain_Comparison.csv');
writetable(displacementComparison,'Lab2_Displacement_Comparison.csv');
save('Lab2_CST_Patch_Test_Results.mat','nodes','elements','D','Kglobal', ...
     'areaAll','B_all','Ke_all','dExact','strainAll','stressAll', ...
     'strainTable','displacementComparison','scaleFactor','deformedNodes');

figHandles = findall(groot,'Type','figure');
for k = 1:length(figHandles)
    figName = get(figHandles(k),'Name');
    if isempty(figName)
        figName = sprintf('Figure_%d',k);
    end
    safeName = regexprep(figName,'[^a-zA-Z0-9_-]','_');
    saveas(figHandles(k),[safeName,'.png']);
end

fprintf('\n=====================================================\n');
fprintf('PATCH TEST COMPLETED. CSV, MAT, AND PNG FILES SAVED.\n');
fprintf('Exact strain: ex = 0.001, ey = 0.002, gxy = 0.\n');
fprintf('Both CST elements should reproduce these strains.\n');
fprintf('=====================================================\n');
```

6.RESULTS

cst_element.m | cst2.m | LAST.m | cstpatch.m

Command Window

New to MATLAB? See resources for Getting Started.

```
=====================================================
            CST PATCH TEST - LAB 2
=====================================================

1) MATERIAL MATRIX D (MPa):
   2.1978e+05        65934            0
        65934   2.1978e+05            0
            0            0        76923


2) NODAL DISPLACEMENTS (prescribed/exact values):
    Node    x_mm    y_mm    u_mm     v_mm
    ____    ____    ____    _____    _____

     1       0       0          0        0
     2       1       0      0.001        0
     3       1       1      0.001    0.002
     4       0       1          0    0.002

Global displacement vector [u1 v1 u2 v2 u3 v3 u4 v4]^T (mm):
           0
           0
       0.001
           0
       0.001
       0.002
           0
       0.002
```

```
-----------------------------------------------------
ELEMENT 1 (Nodes 1-2-3)
Area A1 = 0.500000 mm^2
b coefficients = [-1.0000 1.0000 0.0000]
c coefficients = [0.0000 -1.0000 1.0000]
B matrix for Element 1:
    -1     0     1     0     0     0
     0     0     0    -1     0     1
     0    -1    -1     1     1     0

Element stiffness matrix Ke1 (N/mm):
   1.0989e+05           0  -1.0989e+05       32967           0       -32967
             0       38462       38462      -38462      -38462            0
  -1.0989e+05       38462   1.4835e+05      -71429      -38462        32967
       32967      -38462      -71429   1.4835e+05       38462  -1.0989e+05
             0      -38462      -38462       38462       38462            0
      -32967           0       32967  -1.0989e+05           0   1.0989e+05

Element displacement vector de1 (mm):
           0
           0
       0.001
           0
       0.001
       0.002
```

Element strain [ex ey gxy] for Element 1:

0.001	0.002	0

Element stress [sx sy txy] for Element 1 (MPa):

351.65	505.49	0

ELEMENT 2 (Nodes 1-3-4)

Area A2 = 0.500000 mm^2

b coefficients = [0.0000 1.0000 -1.0000]

c coefficients = [-1.0000 0.0000 1.0000]

B matrix for Element 2:

0	0	1	0	-1	0
0	-1	0	0	0	1
-1	0	0	1	1	-1

Element stiffness matrix Ke2 (N/mm):

38462	0	0	-38462	-38462	38462
0	1.0989e+05	-32967	0	32967	-1.0989e+05
0	-32967	1.0989e+05	0	-1.0989e+05	32967
-38462	0	0	38462	38462	-38462
-38462	32967	-1.0989e+05	38462	1.4835e+05	-71429
38462	-1.0989e+05	32967	-38462	-71429	1.4835e+05

Element displacement vector de2 (mm):

0
0
0.001
0.002
0
0.002

Element strain [ex ey gxy] for Element 2:

0.001	0.002	0

Element stress [sx sy txy] for Element 2 (MPa):

351.65	505.49	0

3) ELEMENT AREAS (mm^2):

Element	Area_mm2
1	0.5
2	0.5

4) ASSEMBLED GLOBAL STIFFNESS MATRIX K (N/mm):

1.4835e+05	0	-1.0989e+05	32967	0	-71429	-38462	38462	
0	1.4835e+05	38462	-38462	-71429	0	32967	-1.0989e+05	
-1.0989e+05	38462	1.4835e+05	-71429	-38462	32967	0	0	
32967	-38462	-71429	1.4835e+05	38462	-1.0989e+05	0	0	
0	-71429	-38462	38462	1.4835e+05	0	-1.0989e+05	32967	
-71429	0	32967	-1.0989e+05	0	1.4835e+05	38462	-38462	
-38462	32967	0	0	-1.0989e+05	38462	1.4835e+05	-71429	
38462	-1.0989e+05	0	0	32967	-38462	-71429	1.4835e+05	

5) EXACT VS CST ELEMENT STRAINS (dimensionless):

Element	Exact_ex	CST_ex	Exact_ey	CST_ey	Exact_gxy	CST_gxy
1	0.001	0.001	0.002	0.002	0	0
2	0.001	0.001	0.002	0.002	0	0

6) EXACT VS CST NODAL DISPLACEMENT COMPARISON (mm):

Node	Exact_u_mm	CST_u_mm	Exact_v_mm	CST_v_mm
1	0	0	0	0
2	0.001	0.001	0	0
3	0.001	0.001	0.002	0.002
4	0	0	0.002	0.002

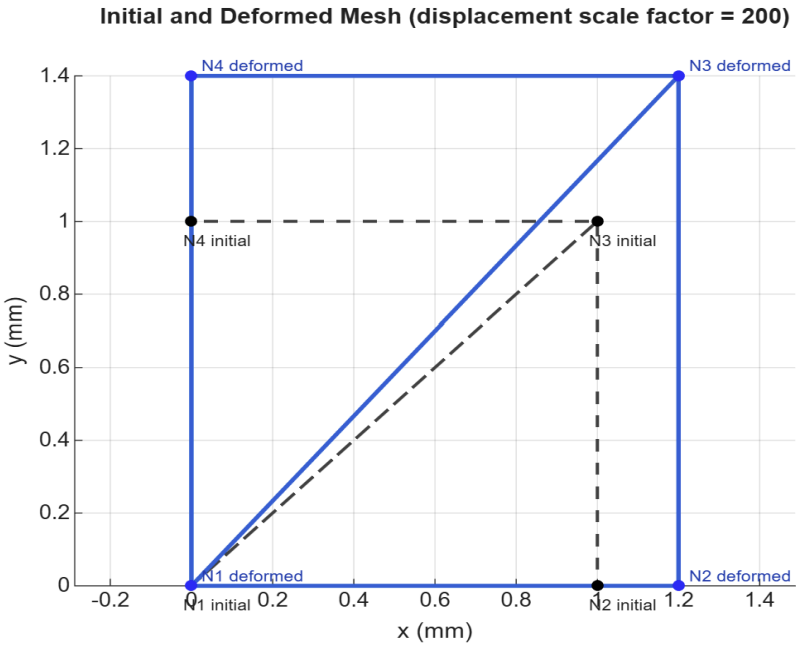
==========

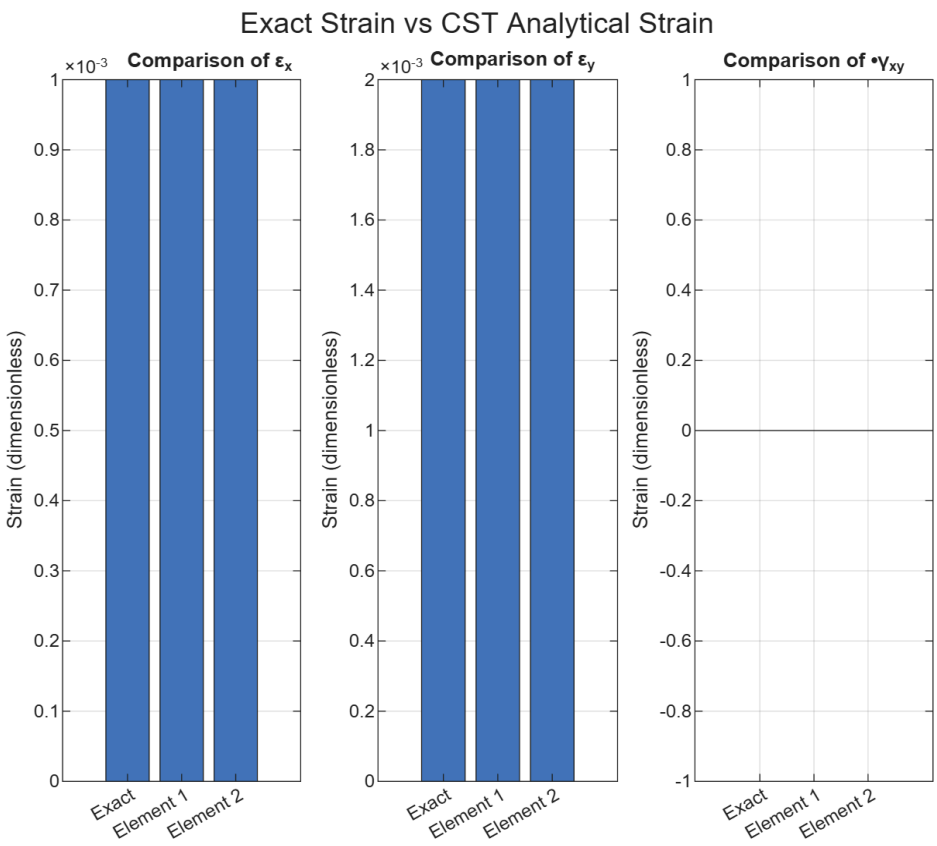
PATCH TEST COMPLETED. CSV, MAT, AND PNG FILES SAVED.

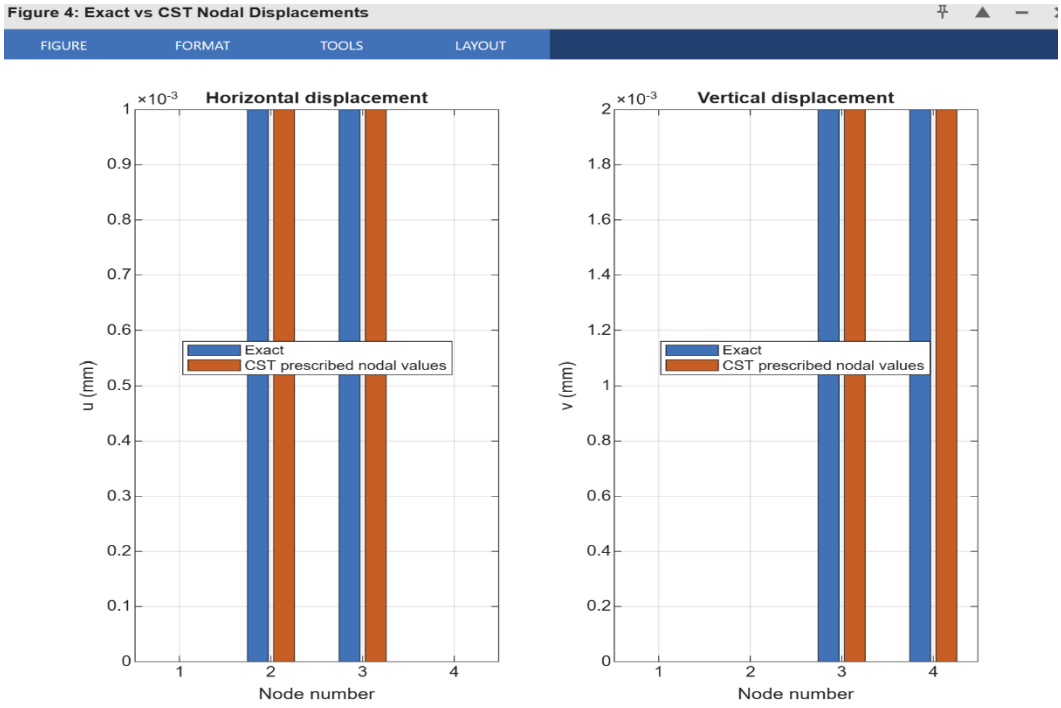
Exact strain: ex = 0.001, ey = 0.002, gxy = 0.

Both CST elements should reproduce these strains.

==========


Initial and Deformed Mesh (displacement scale factor = 200)
N4 deformed
N3 deformed
N4 initial
N3 initial
N1 deformed
N2 deformed
N1 initial
N2 initial
y (mm)
x (mm)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
0.2
0.4
0.6
0.8
1.2
1.4


Exact Strain vs CST Analytical Strain
Comparison of εx
Comparison of εy
Comparison of •γxy
×10-3
Strain (dimensionless)
Exact
Element 1
Element 2



Given:
u = 0.001x
v = 0.002y
The exact strains are calculated as:

- $\varepsilon x = \partial u/\partial x = 0.001$
- $\varepsilon y = \partial v/\partial y = 0.002$
- $\gamma xy = \partial u/\partial y + \partial v/\partial x = 0$

Exact and Simulated Strain Comparison

Element	Exact ε_x	MATLAB ε_x	Exact ε_y	MATLAB ε_y	Exact γ_{xy}	MATLAB γx_y
1	0.001	0.001	0.002	0.002	0	0
2	0.001	0.001	0.002	0.002	0	0

The analytical stresses are calculated using the plane-stress material matrix, D, and the exact strain values. The simulated values below are the MATLAB-computed stresses

Analytical and Simulated Stress Comparison

Element	Stress component	Analytical stress (MPa)	MATLAB stress (MPa)
1	σ_x	351.65	351.65
1	σ_y	505.49	505.49
1	τ_{xy}	0	0
2	σ_x	351.65	351.65
2	σ_y	505.49	505.49
2	τ_{xy}	0	0

Tip Displacement Comparison

Since the CST patch test does not have a physical tip, Node 3 (x = 1 mm, y = 1 mm) is considered the reference corner for displacement comparison.

Displacement parameter	Analytical (mm)	MATLAB (mm)
Horizontal displacement, u_3	0.001	0.001
Vertical displacement, v_3	0.002	0.002
Resultant displacement	0.002236	0.002236

Resultant displacement = $\sqrt{(u_3^2 + v_3^2)} = \sqrt{(0.001^2 + 0.002^2)} = 0.002236$ mm.

Discussion

The MATLAB results agree with the analytical calculations for nodal displacements, strains, and stresses in both CST elements. Each element reproduces the prescribed linear displacement field, yielding $\varepsilon x = 0.001$, $\varepsilon y = 0.002$, and $\gamma xy = 0$. The matching results demonstrate that the CST formulation accurately represents a linear displacement field for the given patch test.

Conclusion

The CST patch test was successfully completed using MATLAB. The computed nodal displacements, strains, and stresses matched the analytical values. This verifies the ability of the CST element to reproduce a linear displacement field for the specified mesh and plane-stress conditions.

Lab 4.3: Cantilever Beam Using CST Elements in MATLAB

1. Problem Statement

A rectangular cantilever beam subjected to a downward tip load of 100 N is to be analysed using the Constant Strain Triangle (CST) finite element method in MATLAB. The beam is fixed at its left edge, while the load is applied at the mid-height node of the free end.

The beam is discretised into triangular elements, and the global stiffness matrix is assembled to calculate the nodal displacements and vertical tip displacement. A mesh refinement study is performed to compare the FEM results with the analytical solution and examine the convergence behaviour of CST elements.

2. Objectives

- To generate a structured mesh of triangular CST elements for a rectangular cantilever beam using MATLAB.
- To calculate element stiffness matrices and assemble the global stiffness matrix.
- To apply the boundary conditions and external load and solve for nodal displacements.
- To determine the vertical displacement at the free-end loading node.
- To compare the FEM tip displacement with the analytical solution obtained using Euler–Bernoulli beam theory.
- To perform a mesh refinement study and investigate the limitations of CST elements in bending-dominated problems.

3. Given Data

Parameter	Symbol	Value
Beam length	L	1000 mm
Beam height	H	100 mm
Thickness	t	1 mm
Young's modulus	E	210,000 MPa
Poisson's ratio	ν	0.3
Applied vertical load	Fy	−100 N
Left-edge boundary condition	—	u = 0, v = 0
Element type	—	CST triangular element
Analysis type	—	Linear static, plane stress

Mesh configurations used:

Mesh	Length segments (N_x)	Height segments (N_y)	Number of triangular elements
1	10	2	40
2	20	4	160
3	40	8	640
4	80	16	2560

4.Analytical Calculations

Given data

- Beam length, L = 1000 mm
- Beam height, H = 100 mm
- Beam thickness, t = 1 mm
- Young's modulus, E = 210,000 MPa = 210,000 N/mm²
- Poisson's ratio, ν = 0.3
- Vertical tip load, Fy = −100 N
- Left edge: fixed, u = 0 and v = 0

(a)Calculate the second moment of area (I)

For a rectangular cross-section,

$$I = \frac{tH^3}{12}$$

Substituting the given values:

$$I = \frac{1(100)^3}{12}$$

$$I = \frac{1{,}000{,}000}{12}$$

$$\boxed{I = 83{,}333.33 \text{ mm}^4}$$

(b)Calculate the analytical tip deflection

According to Euler–Bernoulli beam theory, the tip deflection of a cantilever beam subjected to a point load at its free end is:

$$\delta = \frac{|F_y|L^3}{3EI}$$

Substituting the values:

$$\delta = \frac{100(1000)^3}{3(210000)(83333.33)}$$

Calculate the numerator:

$$100(1000)^3 = 100{,}000{,}000{,}000$$

Calculate the denominator:

$$3(210000)(83333.33) = 52{,}500{,}000{,}000$$

Therefore,

$$\delta = \frac{100{,}000{,}000{,}000}{52{,}500{,}000{,}000}$$

$$\boxed{\delta = 1.90476 \text{ mm}}$$

Since the load acts downward, the signed vertical tip displacement is:

$$\boxed{v_{\text{tip}} = -1.90476 \text{ mm}}$$

(c) Calculate the maximum bending moment

The maximum bending moment occurs at the fixed end of the cantilever.

$$M_{\max} = |F_y|L$$

Substituting:

$$M_{\max} = 100(1000)$$
$$\boxed{M_{\max} = 100{,}000 \text{ N mm}}$$

(d) Calculate the maximum bending stress

For a beam under pure bending, the bending stress is:

$$\sigma_{\max} = \frac{M_{\max} c}{I}$$

where $c = H/2 = 50$mm is the distance from the neutral axis to the outermost surface.

Substituting:

$$\sigma_{\max} = \frac{100000(50)}{83333.33}$$
$$\boxed{\sigma_{\max} = 60 \text{ MPa}}$$

This is the nominal maximum bending stress predicted by elementary beam theory, occurring at the fixed end's outer surface.

(e)Summary of analytical results

Parameter	Analytical value
Second moment of area, I	83,333.33 mm^4
Maximum bending moment	100,000 N mm
Maximum nominal bending stress	60 MPa
Tip deflection magnitude	1.90476 mm
Signed vertical tip displacement	−1.90476 mm

The analytical vertical tip displacement is −1.90476 mm. This value is used as the reference for comparison with the MATLAB CST finite element results for the different mesh refinements.

5. Procedure

- Open MATLAB and create a new script for the cantilever beam analysis.
- Define the beam geometry, material properties, thickness, and applied tip load using consistent N–mm–MPa units.
- Generate the nodal coordinates programmatically using the specified number of segments along the length and height.
- Divide each rectangular mesh cell into two triangular CST elements and establish the element connectivity.
- Calculate the area and strain-displacement matrix (B) for each triangular element.
- Construct the plane-stress material matrix (D) using Young's modulus and Poisson's ratio.
- Calculate the element stiffness matrix using $Ke = t \times A \times B^T \times D \times B$.
- Assemble all element stiffness matrices into the global stiffness matrix.
- Apply the fixed boundary conditions by setting horizontal and vertical displacements to zero at all nodes along the left edge.
- Apply the −100 N vertical load at the mid-height node on the free end.
- Solve the reduced global stiffness equations to determine the nodal displacements and extract the vertical displacement at the loading node.
- Calculate the analytical tip deflection using Euler–Bernoulli beam theory and compare it with the FEM result.
- Repeat the analysis for progressively refined meshes and record the tip displacement and percentage error.
- Plot the initial and deformed meshes and the mesh convergence curve.
- Discuss the convergence behaviour and the limitations of CST elements for bending-dominated problems.

6.MATLAB CODE

```
%% LAB 3: CANTILEVER BEAM USING CST ELEMENTS
% Units: N, mm, MPa
% Plane-stress Constant Strain Triangle (CST) formulation
% Load is applied at the mid-height node of the free end.
% Save as Lab3_Cantilever_CST.m and click Run.

clear; clc; close all;

%% 1. MATERIAL AND GEOMETRY
L = 1000;          % Beam length (mm)
H = 100;           % Beam height (mm)
t = 1;             % Thickness (mm)
E = 210e3;         % Young's modulus (MPa = N/mm^2)
nu = 0.3;          % Poisson's ratio
Fy = -100;         % Downward tip load (N)

D = E/(1-nu^2) * [1, nu, 0;
                  nu, 1, 0;
                  0, 0, (1-nu)/2];

%% 2. ANALYTICAL EULER-BERNOULLI TIP DEFLECTION
I = t*H^3/12;
delta_exact = abs(Fy)*L^3/(3*E*I);

fprintf('=====================================================\n');
fprintf(' LAB 3: CANTILEVER BEAM USING CST ELEMENTS\n');
fprintf('=====================================================\n');
fprintf('Second moment of area I = %.6f mm^4\n', I);
fprintf('Analytical tip deflection magnitude = %.6f mm\n', delta_exact);
fprintf('Signed analytical vertical displacement = %.6f mm\n\n', -delta_exact);

%% 3. MESH REFINEMENT CASES
Nx_list = [10 20 40 80];
Ny_list = [2 4 8 16];  % Even values provide a node at y = H/2
nCases = numel(Nx_list);
tipDisp = zeros(nCases,1);
relError = zeros(nCases,1);
numNodes = zeros(nCases,1);
numElements = zeros(nCases,1);

for icase = 1:nCases
    Nx = Nx_list(icase);
    Ny = Ny_list(icase);

    %% 4. GENERATE STRUCTURED NODES
    xvals = linspace(0,L,Nx+1);
    yvals = linspace(0,H,Ny+1);
    [X,Y] = meshgrid(xvals,yvals);
    nodes = [X(:), Y(:)];
    nodeID = @(ix,iy) iy + 1 + ix*(Ny+1);

    %% 5. SPLIT EACH RECTANGULAR CELL INTO TWO TRIANGLES
```

```
elements = zeros(2*Nx*Ny,3);
e = 0;
for ix = 0:Nx-1
    for iy = 0:Ny-1
        n1 = nodeID(ix,iy);
        n2 = nodeID(ix+1,iy);
        n3 = nodeID(ix+1,iy+1);
        n4 = nodeID(ix,iy+1);
        e = e + 1; elements(e,:) = [n1 n2 n3];
        e = e + 1; elements(e,:) = [n1 n3 n4];
    end
end

nN = size(nodes,1);
nE = size(elements,1);
numNodes(icase) = nN;
numElements(icase) = nE;
nDOF = 2*nN;

%% 6. ASSEMBLE GLOBAL STIFFNESS MATRIX
K = sparse(nDOF,nDOF);
for el = 1:nE
    conn = elements(el,:);
    xy = nodes(conn,:);
    x = xy(:,1); y = xy(:,2);
    twoA = x(1)*(y(2)-y(3)) + x(2)*(y(3)-y(1)) + x(3)*(y(1)-y(2));
    A = abs(twoA)/2;
    if A <= 0, error('Element %d has zero area.',el); end

    b = [y(2)-y(3); y(3)-y(1); y(1)-y(2)];
    c = [x(3)-x(2); x(1)-x(3); x(2)-x(1)];
    B = (1/(2*A))*[b(1) 0 b(2) 0 b(3) 0;
                   0 c(1) 0 c(2) 0 c(3);
                   c(1) b(1) c(2) b(2) c(3) b(3)];
    Ke = t*A*(B'*D*B);

    dofs = zeros(1,6);
    for a = 1:3
        dofs(2*a-1) = 2*conn(a)-1;
        dofs(2*a) = 2*conn(a);
    end
    K(dofs,dofs) = K(dofs,dofs) + Ke;
end

%% 7. APPLY LOAD AND FIXED BOUNDARY CONDITIONS
F = zeros(nDOF,1);
tipNode = nodeID(Nx,Ny/2);
tipVDOF = 2*tipNode;
F(tipVDOF) = Fy;

fixedNodes = arrayfun(@(iy) nodeID(0,iy),0:Ny);
fixedDOFs = reshape([2*fixedNodes-1; 2*fixedNodes],[],1);
allDOFs = (1:nDOF)';
freeDOFs = setdiff(allDOFs,fixedDOFs);
```

```
    %% 8. SOLVE NODAL DISPLACEMENTS
    U = zeros(nDOF,1);
    U(freeDOFs) = K(freeDOFs,freeDOFs)\F(freeDOFs);
    tipDisp(icase) = U(tipVDOF);
    relError(icase) = abs(abs(tipDisp(icase))-delta_exact)/delta_exact*100;

    fprintf('Mesh %d: Nx=%d, Ny=%d | Nodes=%d | Elements=%d | Tip v=%.6f mm |
Error=%.2f%%\n', ...
        icase,Nx,Ny,nN,nE,tipDisp(icase),relError(icase));

    if icase == 1
        nodes_first = nodes; elements_first = elements; U_first = U;
        tipNode_first = tipNode;
    end
    if icase == nCases
        nodes_final = nodes; elements_final = elements; U_final = U;
    end
end

%% 9. PLOT INITIAL AND DEFORMED MESH (COARSE MESH)
scale = 100;
Uxy_first = reshape(U_first,2,[])';
deformed_first = nodes_first + scale*Uxy_first;

figure('Name','Initial CST Mesh');
triplot(elements_first,nodes_first(:,1),nodes_first(:,2),'k-');
axis equal; grid on; xlabel('x (mm)'); ylabel('y (mm)');
title(sprintf('Initial CST Mesh: Nx=%d, Ny=%d',Nx_list(1),Ny_list(1)));

figure('Name','Initial and Deformed Mesh');
triplot(elements_first,nodes_first(:,1),nodes_first(:,2),'k--');
hold on;
triplot(elements_first,deformed_first(:,1),deformed_first(:,2),'r-');
plot(nodes_first(tipNode_first,1),nodes_first(tipNode_first,2),'bo','MarkerFaceColor'
,'b');
axis equal; grid on; xlabel('x (mm)'); ylabel('y (mm)');
title(sprintf('Initial and Deformed Mesh (magnification factor = %g)',scale));
legend('Initial mesh','Deformed mesh (magnified)','Tip load node','Location','best');

%% 10. MESH CONVERGENCE PLOT
figure('Name','Mesh Convergence');
plot(numElements,abs(tipDisp),'o-','LineWidth',1.5);
hold on; yline(delta_exact,'r--','Analytical magnitude');
grid on; xlabel('Number of CST elements');
ylabel('Tip deflection magnitude (mm)');
title('CST Cantilever Mesh Refinement Study');
legend('CST FEM','Euler-Bernoulli analytical','Location','best');

%% 11. SUMMARY TABLE AND SAVED RESULTS
Results = table(Nx_list(:),Ny_list(:),numNodes,numElements,tipDisp,abs(tipDisp), ...
    repmat(delta_exact,nCases,1),relError, ...
    'VariableNames',{'Nx','Ny','Nodes','Elements','FEM_v_tip_mm', ...

'FEM_deflection_magnitude_mm','Analytical_deflection_mm','Relative_error_percent'});
fprintf('\nMESH REFINEMENT SUMMARY:\n');
```

```
disp(Results);

writetable(Results,'Lab3_CST_Mesh_Convergence.csv');
save('Lab3_CST_Results.mat','Results','D','nodes_first','elements_first','U_first',
...

'nodes_final','elements_final','U_final','delta_exact','I','L','H','E','nu','t','Fy')
;

fprintf('\nSaved: Lab3_CST_Mesh_Convergence.csv and Lab3_CST_Results.mat\n');
fprintf('Figures are open; save them using Figure > Save As if needed.\n');
fprintf('Note: CST linear triangles may be overly stiff and converge slowly for
bending.\n');
```

7.RESULTS

Command Window

New to MATLAB? See resources for Getting Started.

```
 LAB 3: CANTILEVER BEAM USING CST ELEMENTS
==================================================
Second moment of area I = 83333.333333 mm^4
Analytical tip deflection magnitude = 1.904762 mm
Signed analytical vertical displacement = -1.904762 mm

Mesh 1: Nx=10, Ny=2 | Nodes=33 | Elements=40 | Tip v=-0.706872 mm | Error=62.89%
Mesh 2: Nx=20, Ny=4 | Nodes=105 | Elements=160 | Tip v=-1.336339 mm | Error=29.84%
Mesh 3: Nx=40, Ny=8 | Nodes=369 | Elements=640 | Tip v=-1.726893 mm | Error=9.34%
Mesh 4: Nx=80, Ny=16 | Nodes=1377 | Elements=2560 | Tip v=-1.864719 mm | Error=2.10%

MESH REFINEMENT SUMMARY:
```

Nx	Ny	Nodes	Elements	FEM_v_tip_mm	FEM_deflection_magnitude_mm	Analytical_deflection_mm	Relative_error_percent
10	2	33	40	-0.70687	0.70687	1.9048	62.889
20	4	105	160	-1.3363	1.3363	1.9048	29.842
40	8	369	640	-1.7269	1.7269	1.9048	9.3381
80	16	1377	2560	-1.8647	1.8647	1.9048	2.1022

```
Saved: Lab3_CST_Mesh_Convergence.csv and Lab3_CST_Results.mat
Figures are open; save them using Figure > Save As if needed.
```

Figure 1: Initial CST Mesh

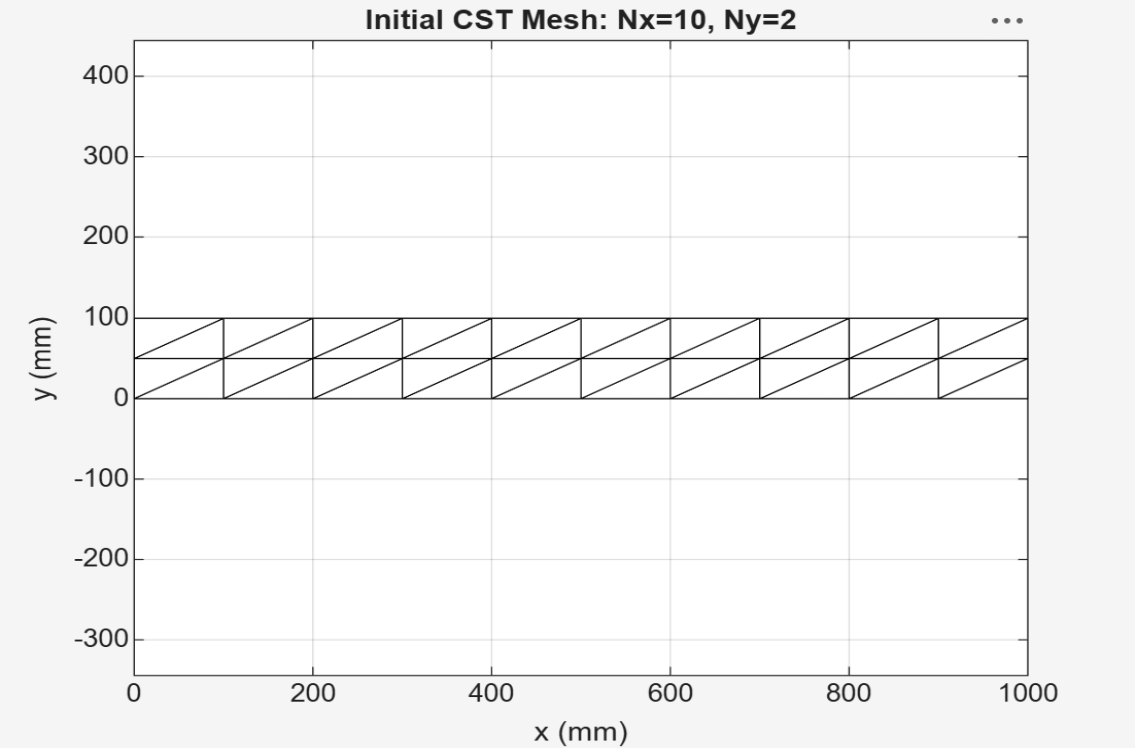
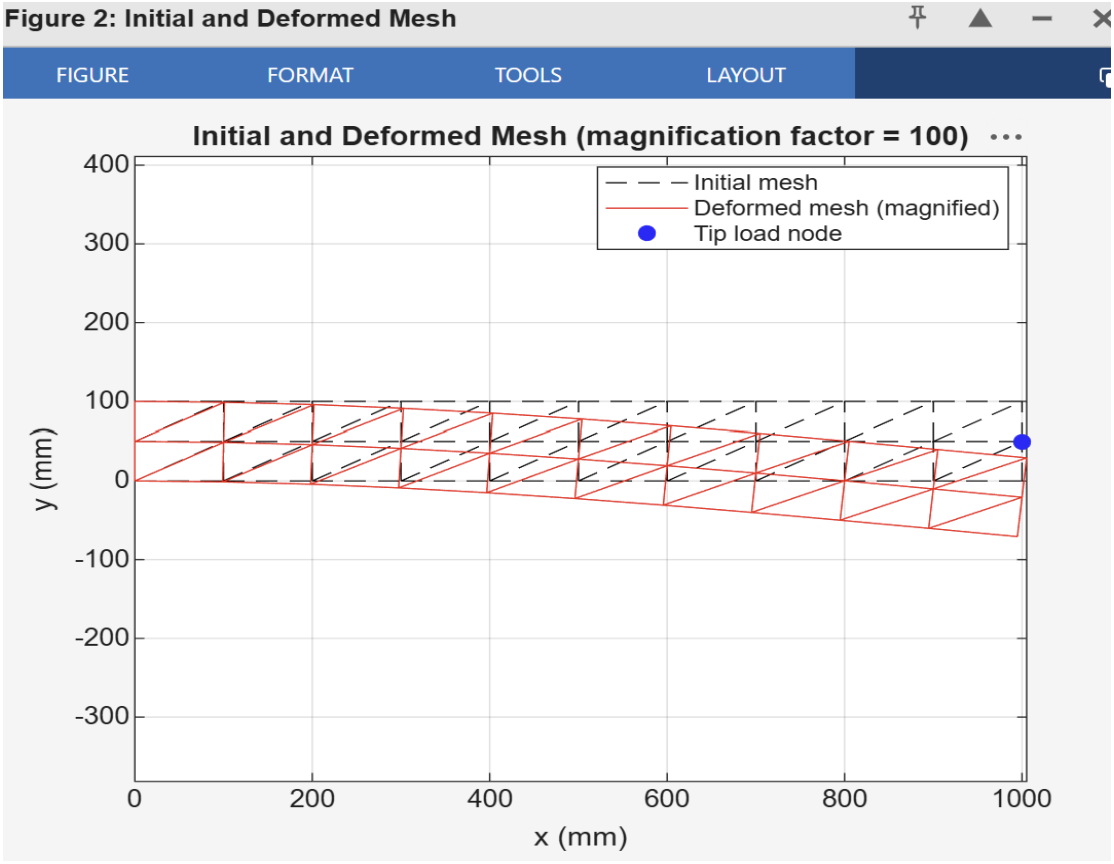
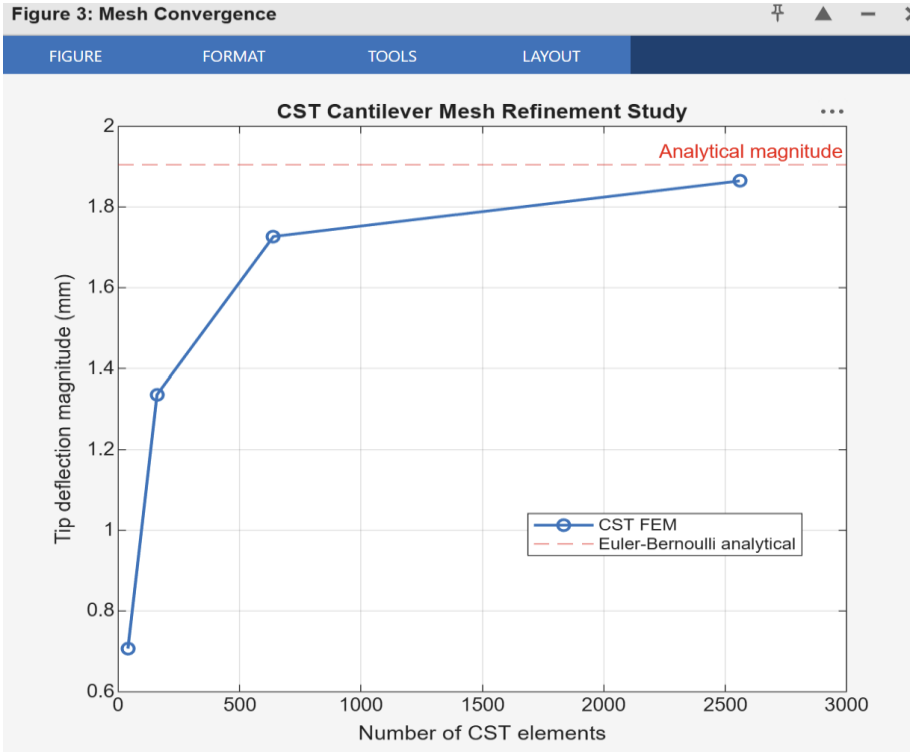


Figure 2: Initial and Deformed Mesh





Comparison of FEM and Analytical Results:

The analytical tip deflection is calculated using the Euler–Bernoulli beam theory:

$$\delta_{\text{analytical}} = \frac{|F_y| L^3}{3EI}$$

Analytical tip deflection magnitude = 1.904762 mm (downward).

Mesh	N_x	N_y	No. of elements	FEM tip displacement (mm)	Analytical displacement (mm)	Relative error (%)
1	10	2	40	−0.706872	−1.904762	62.89
2	20	4	160	−1.336339	−1.904762	29.84
3	40	8	640	−1.726893	−1.904762	9.34
4	80	16	2560	−1.864719	−1.904762	2.10

Relative error is calculated as:

$$\text{Error}(\%) = \frac{\left|\delta_{\text{analytical}} - |\delta_{\text{FEM}}|\right|}{\delta_{\text{analytical}}} \times 100$$

Convergence with mesh refinement:

As the mesh is refined from 40 to 2560 triangular elements, the magnitude of the FEM tip displacement increases from 0.706872 mm to 1.864719 mm, approaching the analytical value of 1.904762 mm. The relative error decreases from 62.89% to 2.10%.

This indicates that the FEM solution converges towards the analytical solution as the mesh becomes finer. The finest mesh provides the closest agreement with the Euler–Bernoulli prediction.

Limitations of CST elements in bending-dominated problems:

- Constant strain assumption: CST elements have constant strain within each triangle and cannot accurately represent the varying strain distribution associated with beam bending using a coarse mesh.
- Overly stiff response: Linear triangular CST elements tend to underestimate beam deflection in bending-dominated problems.
- Slow convergence: A relatively fine mesh is required to obtain accurate results, increasing computational cost.
- Improvement: Mesh refinement improves the prediction. Higher-order triangular elements or beam elements can represent bending more efficiently.

Conclusion

The CST finite element model successfully predicts the cantilever beam's vertical tip displacement. Mesh refinement improves agreement with the analytical solution, reducing the relative error to 2.10% for the finest mesh.

The results demonstrate that CST elements can model bending behaviour, but require sufficient mesh refinement because of their constant-strain formulation.